

Mathematics 2

Homework for Part 1

The solutions are to be submitted as one .zip file to the appropriate mailbox on uclnca. The solutions should contain a .pdf file containing a clear and well described procedure and a code. The homework is worth 2 points. The written exam is worth additional 8 points.

1 Importance sampling

We want to approximate the integral

$$I = \int_0^2 x^{1/3} e^{-x^2} dx.$$

1. Plot the integrand.
2. Let $X \sim \text{Unif}(0, 2)$. The Monte Carlo estimator is $\hat{I}_{\text{MC}} = 2 \cdot \frac{1}{n} \sum_{i=1}^n f(X_i)$. For $n = 10^6$ report the sample mean and sample standard deviation of the 10 estimates.
3. Next consider a proposal density of the form $q(x) = c x^{1/3}$ for $x \in [0, 2]$.
 - (a) Determine the constant c such that q is a probability density.
 - (b) Sample from q using inversion sampling. Compute the CDF of q , invert it explicitly, and use the inverse to generate samples.
 - (c) Estimate the integral using the importance sampling estimator

$$\hat{I}_{\text{IS}} = \frac{1}{n} \sum_{i=1}^n \frac{f(X_i)}{q(X_i)}, \quad X_i \sim q.$$

Use the same procedure as in part 2: $n = 10^6$, repeat 10 times, and report mean and standard deviation.

4. Discuss the following:
 - magnitude of variance reduction;
 - stability and concentration of both estimators;
 - computational cost;
 - suitability of the chosen proposal q ;
 - how the shape of f motivates good choices of importance distributions.

2 Markov chains

Let $(X_n)_{n \in \mathbb{N}}$ be an irreducible Markov chain on the finite state space $S = \{s_1, s_2, s_3, s_4\}$ with the transition matrix

$$K = \begin{pmatrix} 0.1 & 0.6 & 0.3 & 0 \\ 0.2 & 0.3 & 0.4 & 0.1 \\ 0 & 0.5 & 0.3 & 0.2 \\ 0.4 & 0 & 0.4 & 0.2 \end{pmatrix}.$$

1. Show that the chain is irreducible.
2. Find the unique stationary distribution π of K . (You do not need to solve the linear system by hand; you may use software.)
3. For each state i , let $T_i = \min\{t \geq 1 : X_t = i : X_0 = i\}$ be the return time to state i . Show that $E[T_i] = \frac{1}{\pi_i}$.
4. Let $g : S \rightarrow \mathbb{R}$ be defined by $g(1) = 1, g(2) = 0, g(3) = 2, g(4) = 1$. What is the value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n g(X_t)$.
5. Determine whether the chain is aperiodic.

3 Metropolis-Hastings algorithm

Consider the Bayesian model where we observe data $x_1, x_2, \dots, x_n \sim \text{Poisson}(\lambda)$, with unknown rate parameter $\lambda > 0$. Assume the prior distribution for λ is

$$\pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0,$$

that is, a $\text{Gamma}(\alpha, \beta)$ prior. Let the observed data be $x = (4, 1, 3, 5, 2, 4)$.

1. Show that the posterior density of λ is proportional to

$$p(\lambda \mid x) \propto \lambda^{\alpha-1+\sum_{i=1}^n x_i} \exp(-(\beta+n)\lambda).$$

2. Use the proposal distribution

$$\lambda' = \lambda \cdot \exp(\epsilon), \quad \epsilon \sim N(0, \sigma^2).$$

- (a) Write down the acceptance probability

$$A(\lambda, \lambda') = \min\left(1, \frac{p(\lambda' \mid x) q(\lambda \mid \lambda')}{p(\lambda \mid x) q(\lambda' \mid \lambda)}\right).$$

- (b) Choose σ^2 by trial so that the acceptance rate is between 25% and 40%.

3. (a) Generate 4 independent chains, each of length 3000.

- (b) Discard the first 1000 iterations of each chain as burn-in.

Autocorrelation. For an MCMC sample $\{x_t\}_{t=1}^N$, the lag- k autocorrelation is defined by

$$\rho(k) = \frac{\text{Cov}(x_t, x_{t+k})}{\text{Var}(x_t)} = \frac{E[(x_t - \mu)(x_{t+k} - \mu)]}{E[(x_t - \mu)^2]}, \quad k \geq 0,$$

where $\mu = E[x_t]$.

Effective Sample Size (ESS). Given the autocorrelation function $\rho(k)$, the effective sample size is

$$\text{ESS} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho(k)},$$

which measures the number of effectively independent draws contained in N correlated MCMC samples.

- (c) Produce the following diagnostics:

- traceplots of all chains displayed together,
- autocorrelation functions (use in-built function),
- effective sample size (use in-built function).

4. Consider a proposal distribution

$$q(\lambda') = \text{Gamma}(k, x),$$

with (k, x) chosen so that the proposal mean matches the posterior mean from part (1).

- (a) Derive the Metropolis–Hastings acceptance ratio for the independence sampler.
- (b) Compare the following between the two proposal distributions:
- acceptance rates,
 - autocorrelation,
 - ESS,
 - overall sampling efficiency.

Discuss which algorithm performs better and explain why.